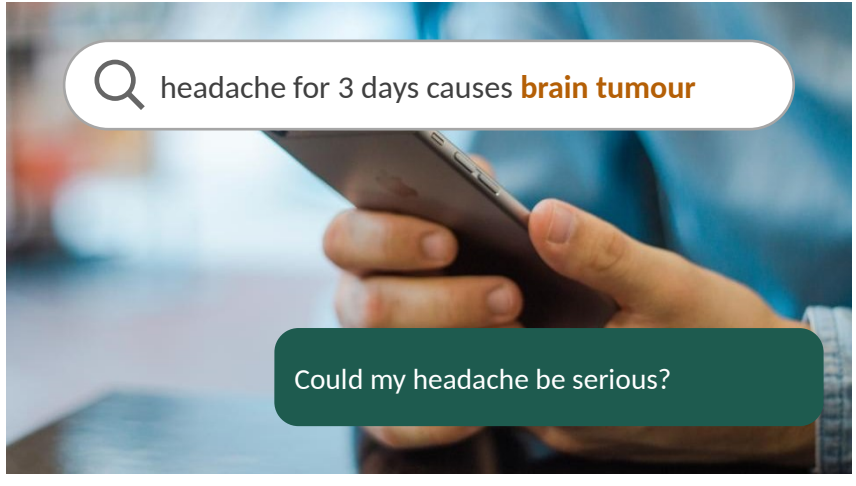
Ministry of Higher Education and Scientific Research
University of Baghdad – College of Medicine

Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad

An online cross-sectional survey

Prepared by:
[Student 1 full name]
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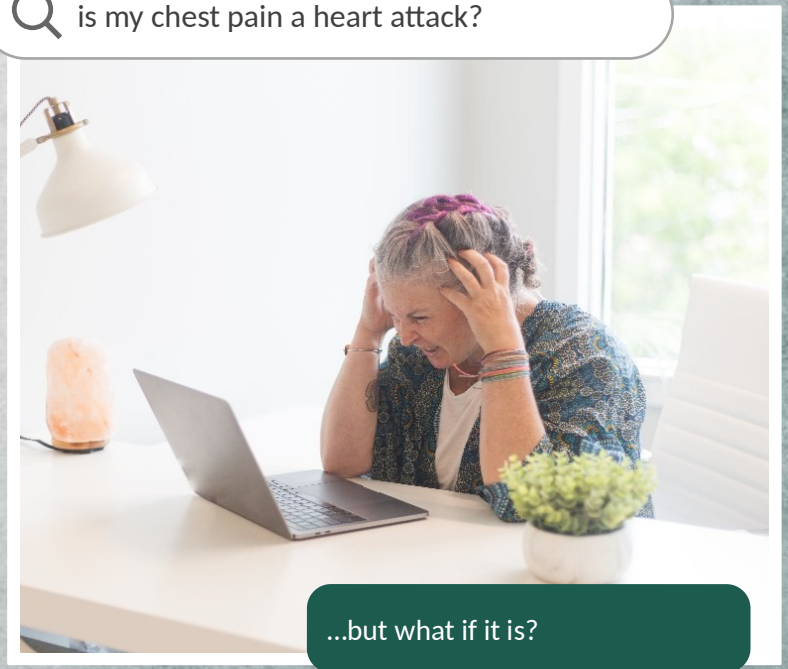


مَا أَنْزَلَ اللَّهُ دَاءً إِلَّا أَنْزَلَ لَهُ شِفَاءً

صحيح البخاري

“He who fears he shall suffer, already suffers what he fears.”

Michel de Montaigne



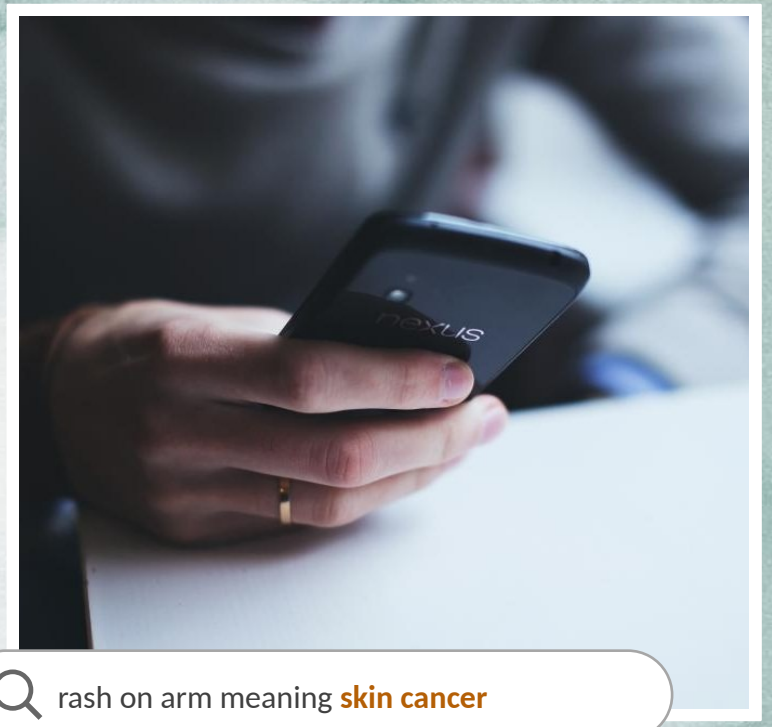
Objectives

- **Describe** online and AI-chatbot health-information seeking and pre-consultation behaviour among adults in Baghdad.
- **Estimate** levels of health anxiety and cyberchondria and the **strength of their association**, independent of other factors.
- **Examine pathways** consistent with a mutually reinforcing cycle between the two.
- **Relate** both to pre-consultation behaviour, **disclosure** of searches to doctors and **trust** in physicians.

Primary hypothesis (H1–H2): health anxiety is positively associated with cyberchondria, independently of socio-demographic and digital-use factors.

Introduction

- Health anxiety
- Cyberchondria
- The vicious cycle
- From search engines to AI chatbots
- Rationale



Health Anxiety

Definition & model

- Persistent preoccupation with **having or acquiring a serious illness**.
- A **continuum**: everyday worry → illness anxiety disorder / somatic symptom disorder (DSM-5).
- Cognitive model: misreading normal sensations → anxiety → **checking & reassurance seeking** → short relief → worry returns.

Burden

- Lifetime prevalence **≈ 6%** in the population, up to **20%** in hospital outpatients.
- Costly: **unnecessary medical contacts**; scores rising over three decades.
- Iraq (COVID-19 online survey, n = 1591): **about half** reported health anxiety.

How we measured it

SHAI-14 (Short Health Anxiety Inventory): 14 items scored 0–3, total **0–42**. Validated in clinical and non-clinical samples; Arabic version validated in Saudi Arabia.

We used **≥ 18** to mark **elevated** health anxiety (a screening threshold, not a diagnosis).

Cyberchondria

Definition: excessive or repeated online searching for health information that **increases, rather than relieves,** distress or anxiety (Starcevic & Berle).

Search logs showed benign-symptom searches often **escalate to rare, serious diagnoses** (White & Horvitz). Searching is normal; it becomes cyberchondria when it is **excessive, compulsive and distressing**.

CSS-12: four facets (3 items each, total 12–60)

Excessiveness	Compulsion	Distress	Reassurance
Searching the same symptom again and again	Searching interrupts work and social life	Feeling worse, panicking after reading	Searching leads to consulting a doctor

Arabic CSS-12 versions show the same four-factor structure (Lebanon, Saudi Arabia).

The Vicious Cycle

- Meta-analyses: health anxiety ↔ cyberchondria **r ≈ 0.62–0.63**.
- High-anxiety users feel **worse** after searching; low-anxiety users feel **relieved**.
- Dutch 4-wave panel (n = 5322): evidence of a **reciprocal** relationship.
- Reassurance gives **short relief**, so the cycle repeats.

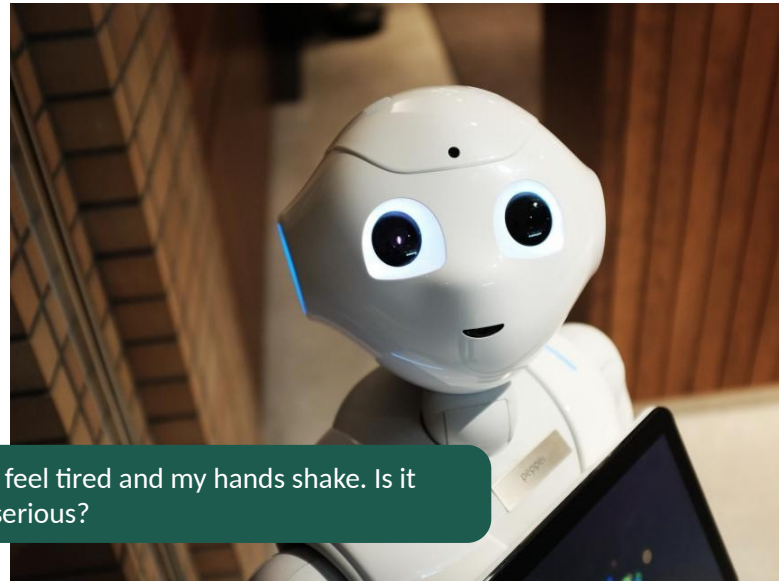


Proposed cycle (cognitive-behavioural model)

From “Dr Google” to “Dr ChatGPT”

What changes with AI chatbots?

- **Fluent, personalised, conversational** answers.
- Can be questioned **again and again**: “one more question”.
- Answers may seem **more empathetic** than physicians’.
- US survey: ~**1 in 5** used ChatGPT for health; ~**1/3** of them changed medication or asked for referrals.



Rationale: first CSS-12 data from Baghdad · AI chatbots studied **separately** from search engines · psychology linked to **real behaviour before the consultation**. More than **80%** of Iraqis are online.

Methodology

- Study design & setting
- Participants & sampling
- Sample size
- Questionnaire
- Statistical analysis

Digital Health Information-Seeking, Health Anxiety, and Cyberchondria Questionnaire
استبيان البحث عن المعلومات الصحية الرقمية، والقلق الصحي، والسايبركوندريا (توهم المرض عبر الإنترنت)

* Indicates required question



1. **By checking the box below, I confirm that I am at least 18 years of age, have read the description of this study, and voluntarily agree to participate. I understand that I can withdraw from this survey at any point without penalty.** *

بالتأشير على المربع أدناه، أؤكد أن عمري لا يقل عن **18** عاماً، وأنني قرأت وصف هذه الدراسة، وأوافق طواعية على المشاركة فيها. كما أفهم أنه يمكنني الانسحاب من هذا الاستبيان في أي وقت دون أي تبعات.

Mark only one oval.

I explicitly agree and consent to participate in this study. / أوافق وأقبل المشاركة في هذه الدراسة البحثية.

Study Design & Settings

Study design & settings

- **Analytical cross-sectional** online survey (STROBE, CHERRIES).
- **Baghdad, Iraq**: 17 July 2026 – 11 September 2026.
- **Convenience + snowball** sampling via Baghdad-based social-media and messaging groups.

Ethics

- **Anonymous & voluntary**; no names or contacts collected.
- **Electronic consent** (and age ≥ 18) required before any question.
- Approval: [committee, number, date].

Criteria	Basis
1. Age ≥ 18 years	Adult population; SHAI & CSS validated in adults
2. Reads Arabic or English	Bilingual questionnaire
3. Gave informed consent	Ethical requirement (e-consent page)
4. Reached through Baghdad channels	Defines the study setting

Sample Size & Questionnaire

Sample size

Primary outcome: **correlation** SHAI ↔ CSS

Fisher's z, two-sided **α = 0.05**, **80% power**

To detect **r = 0.20** → **194** needed

Collected: **n = 300**

Margin of error for %: **±5.7%**

Meta-analytic r ≈ 0.62, so the target is **conservative**.

1. **Consent & socio-demographics**
   age, gender, education, study/work, chronic disease
2. **Digital health seeking**
   tools (search engines, **AI chatbots**…), motives, trust
3. **SHAI-14**
   health anxiety, 0–42
4. **CSS-12**
   cyberchondria, 12–60, four facets
5. **Before the consultation**
   behaviour after searching, **disclosure** to doctor, trust

*Bilingual **Arabic–English** Google Form · Arabic wording by the research team*

Statistical Analysis

Describe

Frequencies, mean ± SD, median (IQR); normality by Shapiro–Wilk

Psychometrics

Cronbach α, McDonald ω, confirmatory factor analysis (CFI, TLI, RMSEA, SRMR)

Associations

Pearson / Spearman (bootstrap CI), partial r; Mann–Whitney, Kruskal–Wallis; χ^2

Multivariable

Hierarchical robust (HC3) regression; logistic & ordinal logistic; AUC

Pathways

Bootstrapped mediation (5000 resamples); moderation by interaction terms

Rigour

Benjamini–Hochberg FDR for multiple testing; sensitivity analyses

Software: **Python 3** (pandas, SciPy, statsmodels, pingouin, semopy), fully scripted · SPSS dataset & syntax provided · **α = 0.05**, two-sided

Results

- Participants
- How people search
- Levels & reliability
- Health anxiety ↔ cyberchondria
- Predictors & pathways
- Behaviour & AI chatbots



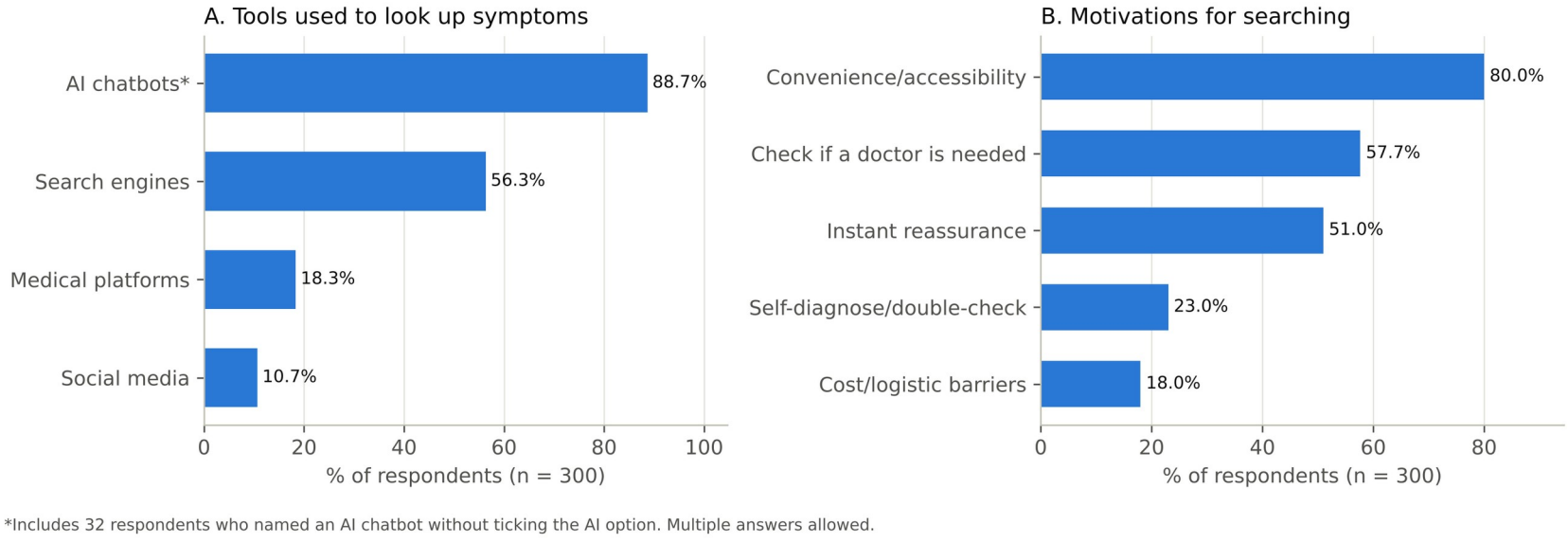
Participants

Table 1: *Socio-demographic characteristics of the participants (n = 300)*

Variable	Category	n (%)
Age group (years)	18–19 / 20–29 / 30–39 / ≥ 40	56 (18.7) / 187 (62.3) / 21 (7.0) / 36 (12.0)
Gender	**Female / Male**	**169 (56.3) / 131 (43.7)**
Education	Currently a student	189 (63.0)
Study / work	**Medical student**	**132 (44.0)**
	Other health-science student / healthcare worker	43 (14.3) / 15 (5.0)
	Non-health student / non-health employee or other	46 (15.3) / 64 (21.3)
Health-related background	Yes	190 (63.3)
Chronic medical condition	Yes	65 (21.7)
Language relied on	Arabic / English	212 (70.7) / 88 (29.3)

Health-related background = medical students, other health-science students and healthcare workers.

How People Search



*Includes 32 respondents who named an AI chatbot without ticking the AI option. Multiple answers allowed.

Figure 3: *Tools used and motivations for searching for health information online (n = 300)*

78.0% ticked AI chatbots

56.3% search engines

41.0% AI without search engine

59.3% use ChatGPT

Levels & Reliability

Table 3: *Descriptive statistics, reliability and language equivalence of the SHAI-14 and CSS-12 (n = 300)*

Scale (range)	Mean ± SD	Median (IQR)	α (95% CI)	ω	α Arabic / English
SHAI-14 (0–42)	**12.35 ± 5.83**	**12 (8–16)**	**0.81 (0.78–0.84)**	**0.82**	**0.81 / 0.81**
CSS-12 (12–60)	**28.86 ± 8.36**	**29 (23–34)**	**0.86 (0.84–0.89)**	**0.87**	**0.86 / 0.88**
Excessiveness (3–15)	9.66 ± 2.94	10 (8–12)	0.77 (0.72–0.81)	0.77	
Compulsion (3–15)	5.00 ± 2.58	4 (3–6)	0.88 (0.86–0.90)	0.89	
Distress (3–15)	7.24 ± 2.74	7 (5–9)	0.76 (0.71–0.81)	0.77	
Reassurance (3–15)	6.96 ± 2.81	7 (5–9)	0.76 (0.71–0.81)	0.77	

CSS-12 four-factor CFA (ML): CFI 0.95, RMSEA 0.070, SRMR 0.053. Arabic vs English: means equal; CSS-12 scalar-invariant, SHAI-14 metric-invariant (multi-group CFA); no item DIF after FDR.

19% elevated health anxiety (SHAI ≥ 18)

15.0–23.8% 95% CI

Health Anxiety ↔ Cyberchondria

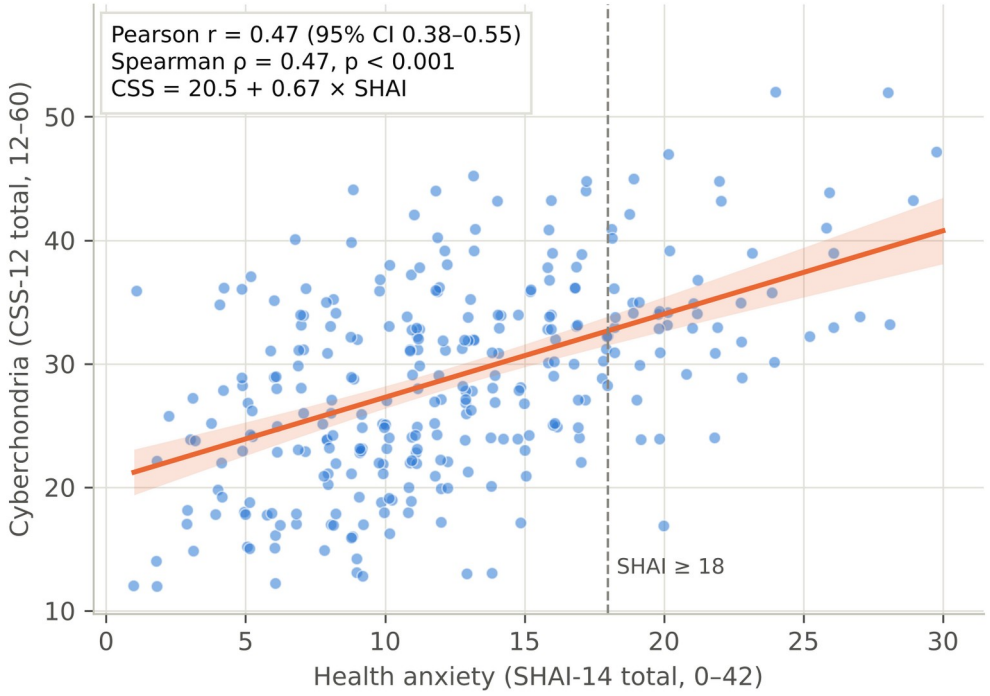


Figure 5: *Health anxiety (SHAI-14) against cyberchondria (CSS-12), least-squares line and 95% band*

r = 0.47

95% CI 0.38–0.55 · p < 0.001

Partial r (all covariates): **0.48**

By CSS facet:

- Excessiveness **0.46**
- Distress **0.45**
- Compulsion 0.25
- Reassurance 0.24

H1 & H2 **supported**

What Predicts Cyberchondria?

Table 9: *Hierarchical robust (HC3) linear regression of cyberchondria (CSS-12 total)*

Block	R^2	ΔR^2	p (change)
1. Socio-demographics	0.030	0.030	—
2. + Digital-use variables	0.133	0.103	< 0.001
3. + Health anxiety (SHAI-14)	**0.317**	**0.184**	**< 0.001**

Final model: *key predictors*

Predictor	B (95% CI)	β	p
SHAI-14 (per point)	**0.68 (0.53–0.84)**	**0.48**	**< 0.001**
AI-chatbot user*	5.33 (2.34–8.32)	0.20	< 0.001
Age 30–39 (vs 20–29)	6.27 (0.19–12.34)	0.19	0.043
Gender, chronic, language, job	n.s.		> 0.5

**Inclusive definition; with the strict definition the AI effect is not significant (slide 23).*

18.4%

extra variance explained by health anxiety, after everything else

3%

explained by demographics alone

Item by Item: What Drives Searching?

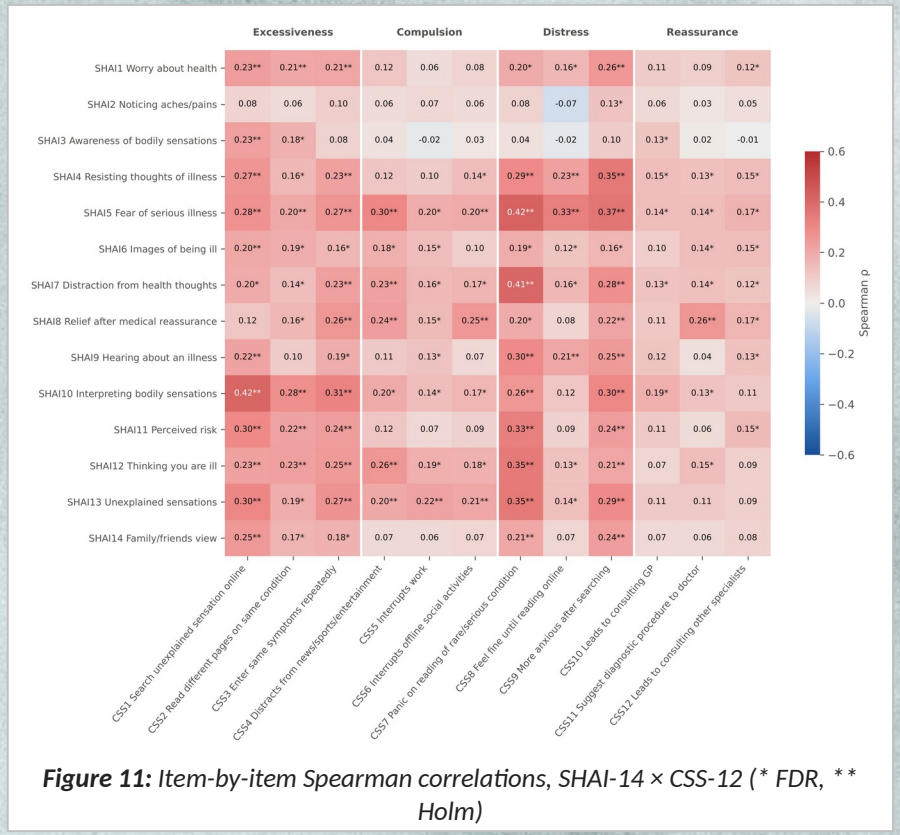


Figure 11: *Item-by-item Spearman correlations, SHAI-14 × CSS-12 (* FDR, ** Holm)*

ρ = 0.43

Fear of serious illness ↔ panic on reading of a rare condition

ρ = 0.42

Interpreting sensations ↔ searching every sensation

≈ 0

Merely noticing aches / sensations

111/168 pairs significant after FDR

Who Has Elevated Health Anxiety?

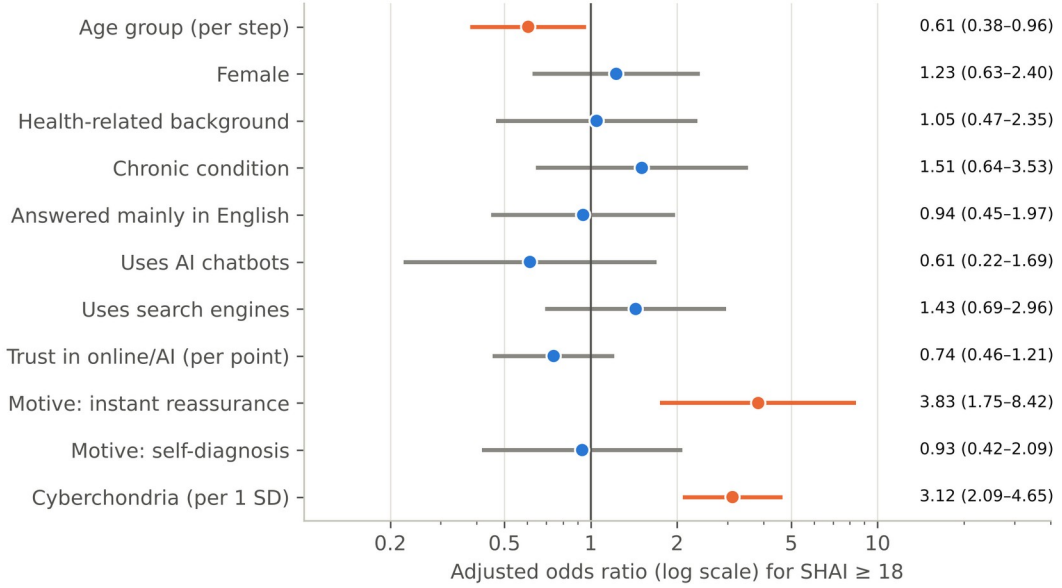


Figure 8: *Adjusted odds ratios for elevated health anxiety (SHAI ≥ 18). Orange: $p < 0.05$*

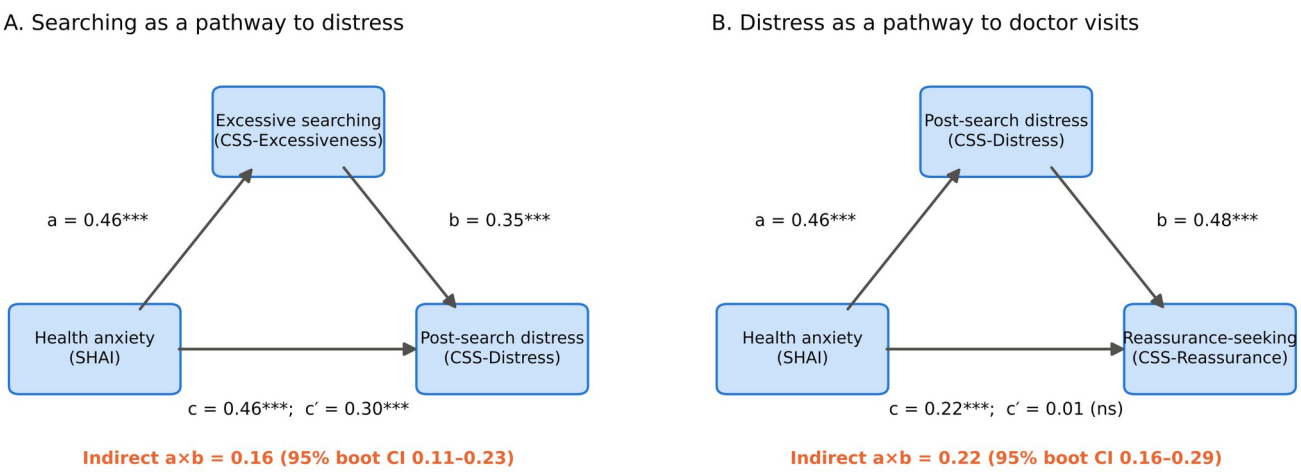
3.12×
odds per SD of cyberchondria

3.83×
if searching for instant reassurance

0.61×
per older age group (protective)

0.83 → 0.79
AUC (apparent → optimism-corrected)

Pathways: Is There a Cycle?



Standardised coefficients adjusted for age, gender, health background, chronic illness, trust and AI use; 5,000 bootstrap resamples. * p<0.05, ** p<0.01, *** p<0.001.

Figure 9: *Mediation models (standardised; 5000 bootstrap resamples)*

Indirect via excessive searching **0.16** (CI 0.11–0.23) · via distress **0.22** (direct → **0.01**) · reverse path also **0.15** → **compatible with a cycle, not proof of direction**

Behaviour Before the Consultation



Figure 4: *Behaviour after searching, disclosure to physicians and effect on the doctor–patient relationship*

34.7%
delay / skip care because reassured

31.7%
never tell their doctor (fear of judgement)

OR 0.57
higher cyberchondria → more disclosure (q = 0.012)

34.7%
say searching strained trust in doctors

AI Chatbots: An Independent Effect?

Table: *Adjusted difference in cyberchondria (CSS-12) for AI-chatbot users*

Definition of AI user	B (CSS points)	95% CI	p
Inclusive: ticked OR named a chatbot	+5.3	2.3–8.3	< 0.001
Strict: ticked "AI chatbots" only	**+1.5**	**—**	**0.31**

32 respondents named a chatbot without ticking it (mean CSS 31.6 vs 24.0 in non-users); only 34 were non-users.

The classification decides it

32 "named but not ticked" respondents had the **highest** scores

Small comparison group

Only **34** true non-users

No moderation

AI use did **not** change the SHAI ↔ CSS link

Conclusion: no reliable evidence that AI use is linked to cyberchondria **beyond health anxiety**.

Discussion

- Our main finding vs the world
- Why weaker than meta-analyses?
- AI chatbots
- The consultation
- Medical students
- Limitations & future directions



So, is searching the problem, or the anxiety?

To our knowledge, this is the **first** study to apply the CSS-12 in a **Baghdad**-based sample, and among the first to place **AI chatbots** alongside search engines as a source of health information before consulting a doctor.

Health anxiety and cyberchondria were **moderately and robustly associated (r = 0.47)**, independent of all other measured factors.



$n = 300 \cdot \mathbf{r = 0.47}$

Comparison with Worldwide Results



Comparison with Worldwide Results

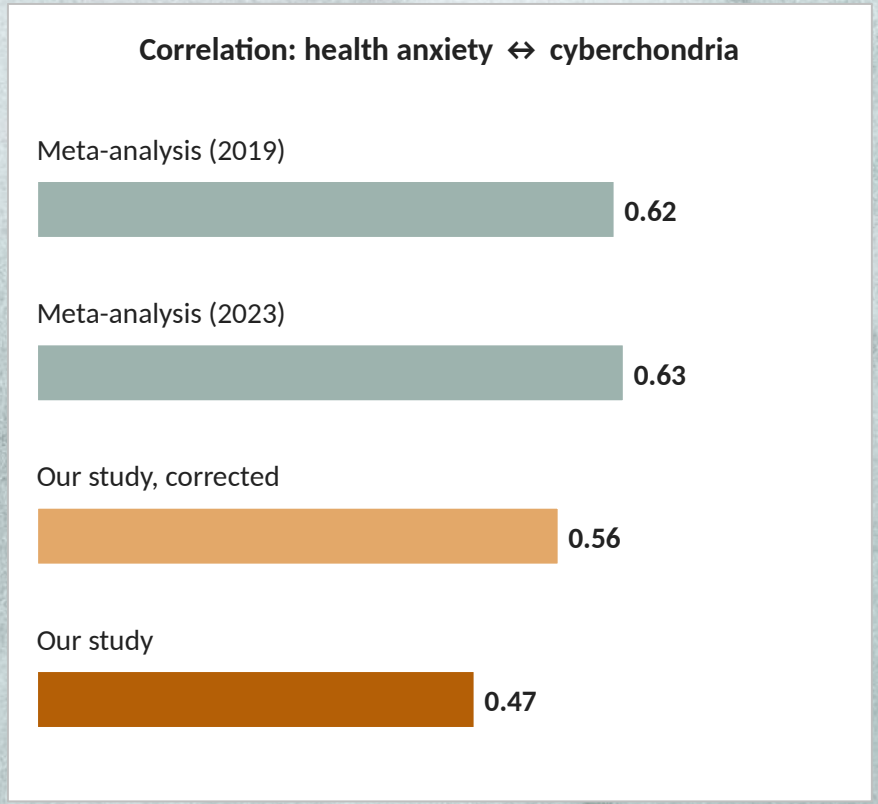
Our study: **r = 0.47**

Meta-analyses (McMullan 2019; Mestre-Bach 2023): **r ≈ 0.62–0.63**

Our r corrected for unreliability: **≈ 0.56**

Why weaker?

1. **Measurement error:** Corrected for reliability, r ≈ 0.56
2. **Narrow range:** Low health anxiety (mean SHAI 12.3) attenuates r
3. **Who answered:** Online recruitment favours confident, habitual users
4. **Short form:** Most earlier studies used the 33-item CSS



Comparison with Regional Studies



Saudi Arabia – Jeddah

Cyberchondria **common**, linked with **smartphone addiction**

Saudi Arabia – primary care

Cyberchondria linked with **health anxiety**

Lebanon

Arabic CSS-12: **same four-factor structure** as ours

Egypt – 1435 medical students

85.8% moderate cyberchondria; health worry predicted it

Our results are **in line** with emerging Arab-region data.

AI Chatbots & the Consultation

AI chatbots

- Our use **78.0%–88.7%** vs ~**20%** (US 2024) and **21%** (2025 survey).
- Young sample: AI use fell from **96%** (18–19 y) to **72%** (≥ 40 y).
- Conversational AI may invite "**one more question**" (excessiveness, reassurance).
- Higher trust → **fewer** multiple-doctor visits, but a trend to **delay care** (OR 1.33, p = 0.088).

The consultation

- Delay because reassured (**34.7%**): saves visits, but risks **delayed diagnosis**; seen at **all** anxiety levels.
- **31.7%** never disclose searches (fear of judgement).
- Higher cyberchondria → **more** disclosure (robust); health anxiety → concealment (**exploratory**).
- H4 only **partially** supported.

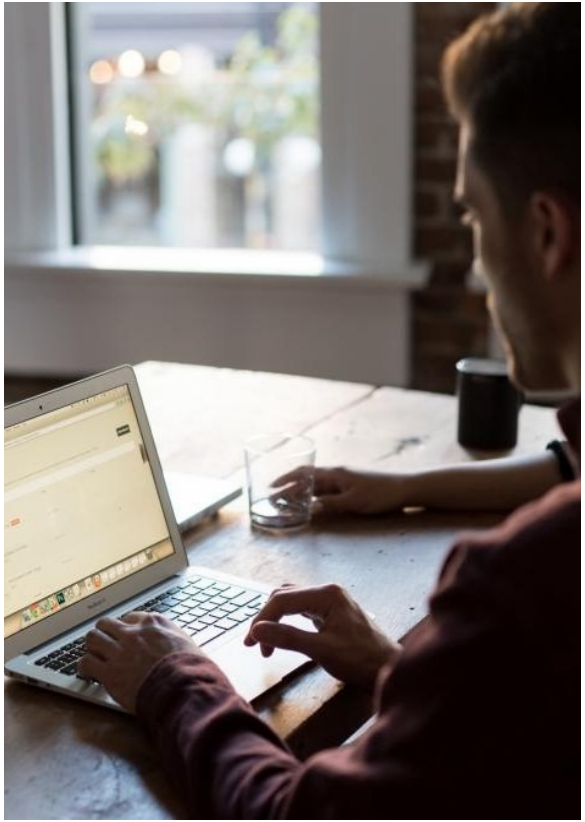
Clinical message: ask every patient, without judgement, what they searched or asked an AI.

44.0% of our sample were **medical students**, a distinct group:

- **Heightened medical awareness** of rare and serious diseases.
- **High academic stress.**
- The popular idea of “medical student syndrome” (weak evidence).

Higher unadjusted health anxiety, but this **disappeared after adjustment**: it reflected their **younger age**.

Cyberchondria scores were **identical** to other participants’: **medical knowledge did not protect** against cyberchondria.



Limitations

1. **Cross-sectional design:** Cannot establish direction; mediation is **statistical, not causal**

2. **Online convenience sample:** Young, **44.0%** medical students; percentages **not generalisable**, associations consistent across subgroups

3. **Arabic wording:** Team translation without formal back-translation; but CSS-12 **fully invariant** and SHAI-14 metric invariant across languages, no item DIF

4. **Single-item measures:** Behaviour, disclosure and trust items **not validated**; labelled exploratory

5. **Multiple testing & self-report:** FDR correction applied; recall and social-desirability bias possible

Future Directions & Recommendations

Future research should prioritise:

- **Longitudinal** / experience-sampling studies of the cycle.
- **Experiments**: AI chatbot vs search-engine answers and anxiety.
- **Probability sampling** of Baghdad adults, including older and less-educated people.
- **Formally adapted Arabic** scales with invariance testing.
- **Intervention trials**: only **9** cyberchondria studies on ClinicalTrials.gov, **none in Iraq**.

Clinicians

Ask, without judgement, what patients searched or asked an AI; screen with the SHAI

Ministry of Health

May consider Arabic digital and AI health-literacy campaigns; warn about false reassurance

AI developers

May consider anxiety-aware answers that do not escalate rare diagnoses

Conclusion

- **AI chatbots** (mainly ChatGPT) are a leading source of health information: **78.0%** ticked them (**88.7%** incl. named), more than search engines.
- **19%** had elevated health anxiety (SHAI ≥ 18, a screening threshold); younger people and reassurance-seekers most affected.
- Health anxiety and cyberchondria are **moderately and robustly associated (r = 0.47)**, mainly through excessive searching and distress, compatible with a **cycle**.
- **No reliable independent effect of AI** beyond health anxiety.
- Searching shapes the consultation: **34.7%** delay care when reassured, **31.7%** never disclose searches.

r = 0.47

Health anxiety and cyberchondria travel together.

Thank you — questions are welcome